

EVERDURE CALTITE SYSTEM

QUALITY ASSURANCE / QUALITY CONTROL

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Appendix A Pro-Forma Order Form

Appendix B Quality Control Forms

Objective

To achieve a watertight, durable reinforced concrete structure by optimising the performance and use of the Everdure Caltite System.

Works Quality Plans

A) Scope of Work

Cementaid (HK) Ltd. (CA) is responsible for the supply and delivery of Caltite waterproofing ingredient as required for the Everdure Caltite System to the Main Contractor's site/Concrete Supplier and carry out dosing and monitoring services required to achieve the stated objective.

CA will advise the Main Contractor on the optimum use of the Everdure Caltite System for the project. During initial design sessions with the Main Contractor, CA will fully advise the Main Contractor of the minimum requirements as documented in our Technical Specification to ensure complete compatibility with the Everdure Caltite System.

Construction techniques and joints details will be discussed with final details approved by C.A.

Optimum design concrete mixes will be determined and trial mixes conducted to ensure concrete performance requirements are met. Trial mixes will be conducted using mix designs are provided by the Main Contractor. If a mix design is deemed to be unsuitable by Cementaid, then Cementaid will request the Main Contractor to provide other mix design(s) for testing.

B) QA Representative

CAHK will pass to the project-in-charge and provide our service if necessary.

C) Material Control

The manufacturing of chemicals for the Everdure Caltite System are covered by CA's manufacturing QA/QC methods which covers quality assurance of raw materials, batching records and final testing of the completed product. All drums are clearly labelled with the drum's contents and batch number.

The Everdure Caltite System consists of one concrete additive - Caltite. Materials are delivered to the Main Contractor nominated site on the day or evening prior to the concrete pour. The quantity delivered is determined from the Main Contractor's order form (refer Appendix A) as faxed to our office plus a contingency amount of approximately 10%. Should this be insufficient for the required pour, then the Main Contractor must allow sufficient time for CA to mobilise extra materials from our factory located in the Fanling district at no penalty to CA.

The ingredients are volume measured at the site and administered into the nominated concrete truck using pumps and dispensers. Delivery dockets are marked as containing the Everdure Caltite System ingredients by the plant monitor. Caltite is dosed at 30 litre/m³, the exact dosage being determined by the trial mixes, and site conditions.

D) Control of Inspection, Measuring and Test Equipment

Quality control testing consists of slump testing and sampling. Sampling is conducted by the site monitor for in-house purposes only. The Main Contractor may wish to take their own test samples as deemed necessary.

Compression testing conducted by a HOKLAS approved laboratory, Cementaid in-house absorption testing is conducted by CA staff. All Cementaid test results remain the property of Cementaid (HK) Ltd. and may be released at the discretion of its Directors.

PVC tube cylinders are used for making in-house samples for Absorption testing in accordance to BS1881: Part 122, 1983, except that testing shall be at age 7 days.

E) Quality Records

Quality control records taken include :

- Appendix A –
Main Contractor's order form
- Appendix B -
 - Pre-Pouring Inspection Form
 - Monitor Check List (Site & Plant)
 - Monitor Daily Worksheet

Copies of these forms can be located in the Appendix A&B.

F) Method Statement for Caltite Concrete Pouring

I. Before the pouring day :

1. Before the day of pouring of **Everdure Caltite System** concrete, Cementaid liaises with the batching plant manager for the appointed concrete supplier, to arrange space for storage of **Everdure Caltite System** materials.
2. Cementaid Monitor visits the site to ensure that all preparation work (eg. installation of waterstops, etc.) is suitable for placing of **Everdure Caltite System** concrete.

II. On the pouring day :

1. Cementaid delivers sufficient **Everdure Caltite System** materials for that day's pour, and appropriate dosing equipment, to the assigned batching plant, at least 1/2 hour before batching starts.
2. Cementaid staff and Monitor at the batching plant dose the correct quantity of the **Everdure Caltite System** ingredients into the mixing truck.
3. Cementaid on-site Monitor checks that the **Everdure Caltite System** concrete delivered to site is of good quality, and advises the Main Contractor on placing Caltite concrete with good concreting practice.

III. After-works service :

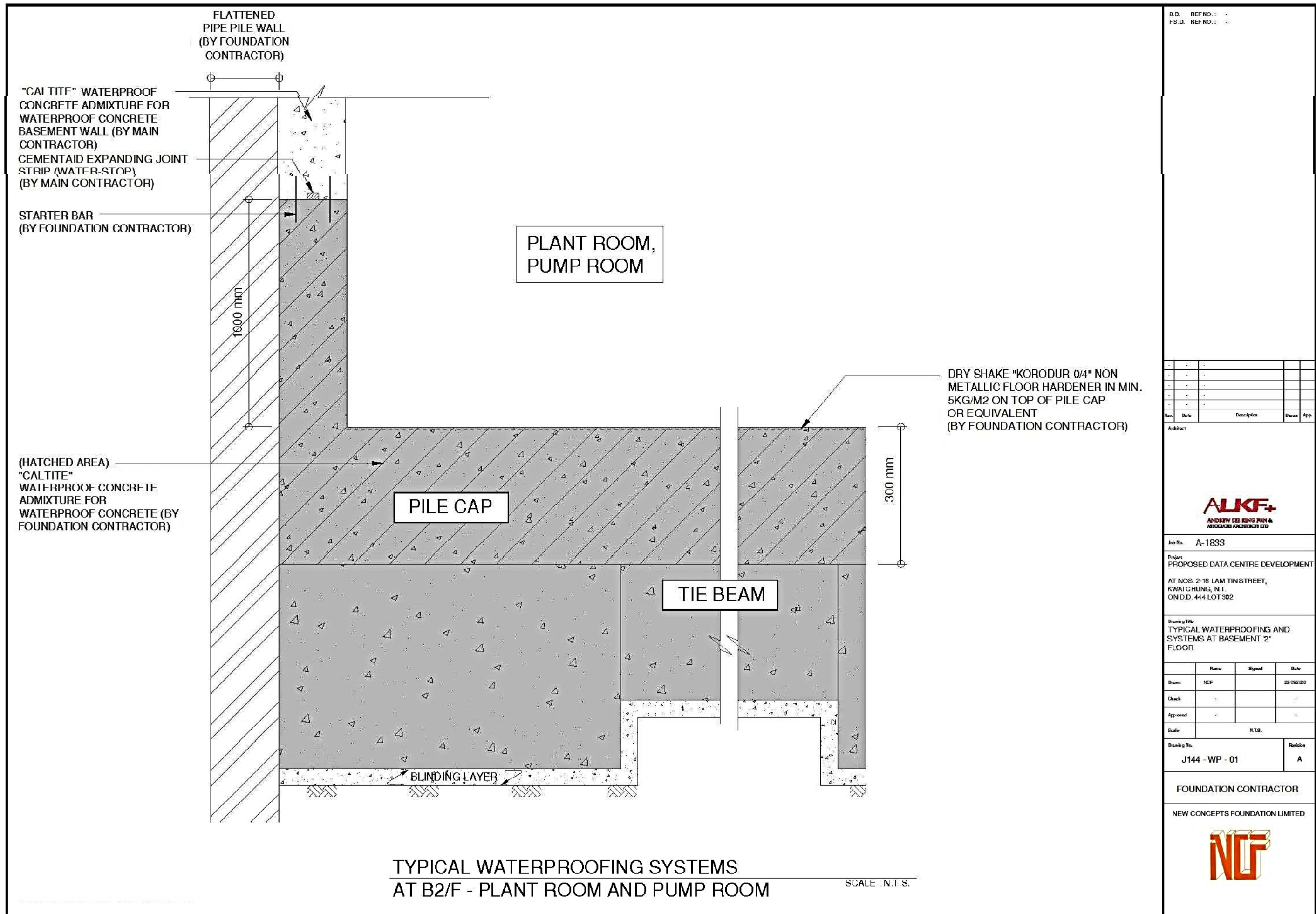
1. Cementaid staff inspect the Caltite concrete after stripping, and locate and identify the cause of any leakage, eg. honeycomb, shrinkage crack, tie hole, embedded steel, etc.
2. Upon the Consultant's approval, Cementaid carries out repairing and grouting works, upon agreement with the Main Contractor on charges, etc.
3. Should any leakage be found to be the result of concreting defect, ie. honeycomb, crack, Cementaid charges for such repairing to the Contractor/concretor.

G) Non Conformance Control

Fresh concrete non conformance normally occurs when the concrete slump is outside the specified maximum limits. This can occur when the batch water added by the batching plant is too high or insufficient allowance has been made for the aggregate moisture.

- . If the workability is less than specified, Superplastet F non-retarding superplasticiser may be added. Should the addition of Superplastet "F" cause the slump to be outside the specified maximum, this concrete is still suitable for placing according to the Cementaid on-site monitor's decision, as the slump has been obtained without raising the W/C ratio.

If the workability is greater than the maximum specified, the truck is to be rejected and no modification in the truck is to be allowed. A record will be kept of the truck number on the Monitor's report to prevent resubmission of the rejected concrete.



B.D. REF NO.:
F.S.D. REF NO.:

Rev.	By	Description	Drawn	App.

Architect



Job No. A-1833

Project PROPOSED DATA CENTRE DEVELOPMENT

AT NOS. 2-16 LAM TIN STREET,
KWAI CHUNG, N.T.
ON D.D. 444 LOT 302

Drawing Title
TYPICAL WATERPROOFING AND
SYSTEMS AT BASEMENT 2'
FLOOR

	Name	Signed	Date
Drawn	NCF		23/09/20
Check			
Approved			

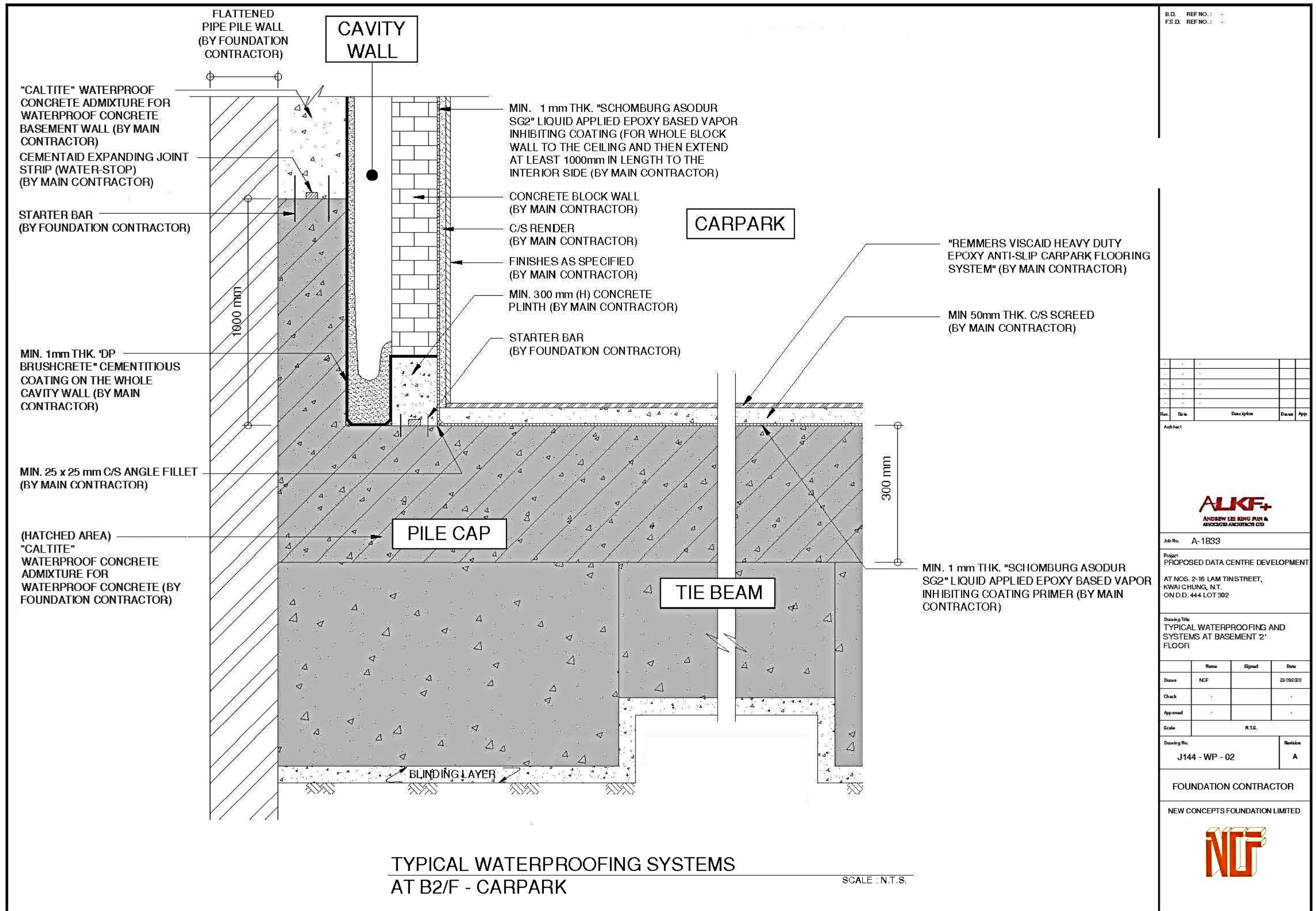
Scale N.T.S.

Drawing No. J144 - WP - 01
Revision A

FOUNDATION CONTRACTOR

NEW CONCEPTS FOUNDATION LIMITED





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Approved			
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Drawing No. J144 - WP - 02
Revision A

FOUNDATION CONTRACTOR

NEW CONCEPTS FOUNDATION LIMITED



**Suggested Construction Joint Details for
Cementaid Everdure Caltite System / 3CC System
Concrete Structures**

INTRODUCTION

No water or moisture will transpire through properly compacted CEMENTAID Everdure Caltite System / 3CC System concrete. Therefore, only joints, or holes through Everdure Caltite System / 3CC System structure need to be sealed. The following suggested procedure have been used in thousands of various Everdure Caltite System / 3CC System structures internationally since 1960, and are proven effective, reliable and maintenance-free over this period. They are very simple and easy to handle, purposely so. The materials described are available from your CEMENTAID representative to check the detail drawings before tender stage and to be present on site to assist the contractor in achieving optimum results.

SPECIFICATION

All construction joints, or projections through CEMENTAID Everdure Caltite System / 3CC System concrete are to be constructed in accordance with details supplied by CEMENTAID LTD unless otherwise shown on the working drawings.

Fig A : Roof / Podium Slab joints Detail

1. Form material to be treated with CEMENTAID CALSTRIP form release and keying agent.
2. On removal of joint form, surface to be washed and brushed clean of all loose material or retarded mortar.
3. CEMENTAID EXPANDING JOINT STRIP to be applied to the horizontal face below the top reinforcement mat of the step joint.
4. Prior to continuing the pour, the step joint is to liberally and thoroughly brushed with a CEMENTAID WELDCRETE Bonding Agent / cement slurry (1:2).
5. New concrete is to be placed against treated prepared step joint and properly compacted against the wet WELDCRETE.

NOTE : WELDCRETE must be wet for optimum bond, do not allow WELDCRETE to dry or become tacky or bond failure will occur. In case of drying out, apply fresh WELDCRETE slurry before continuing.

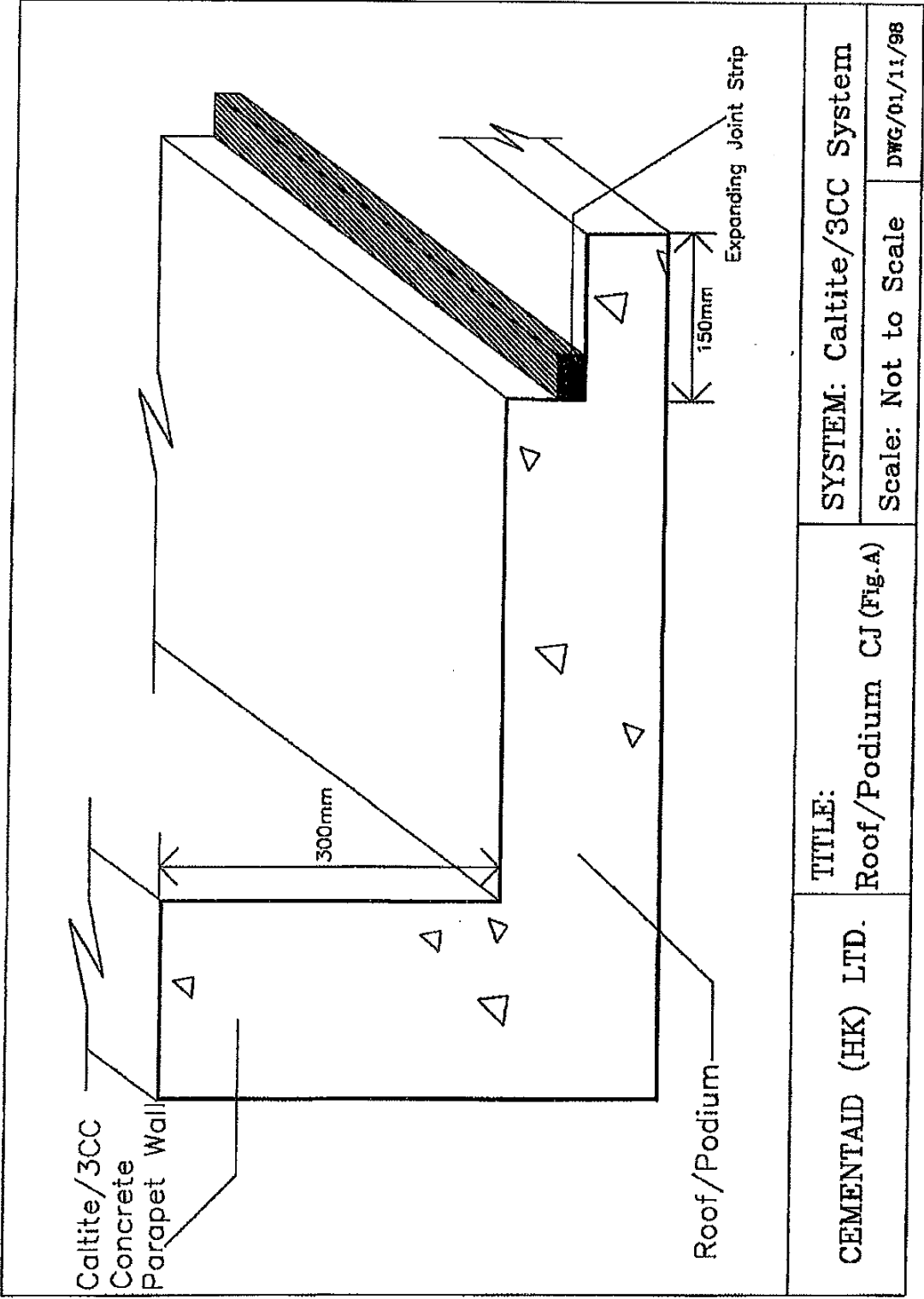
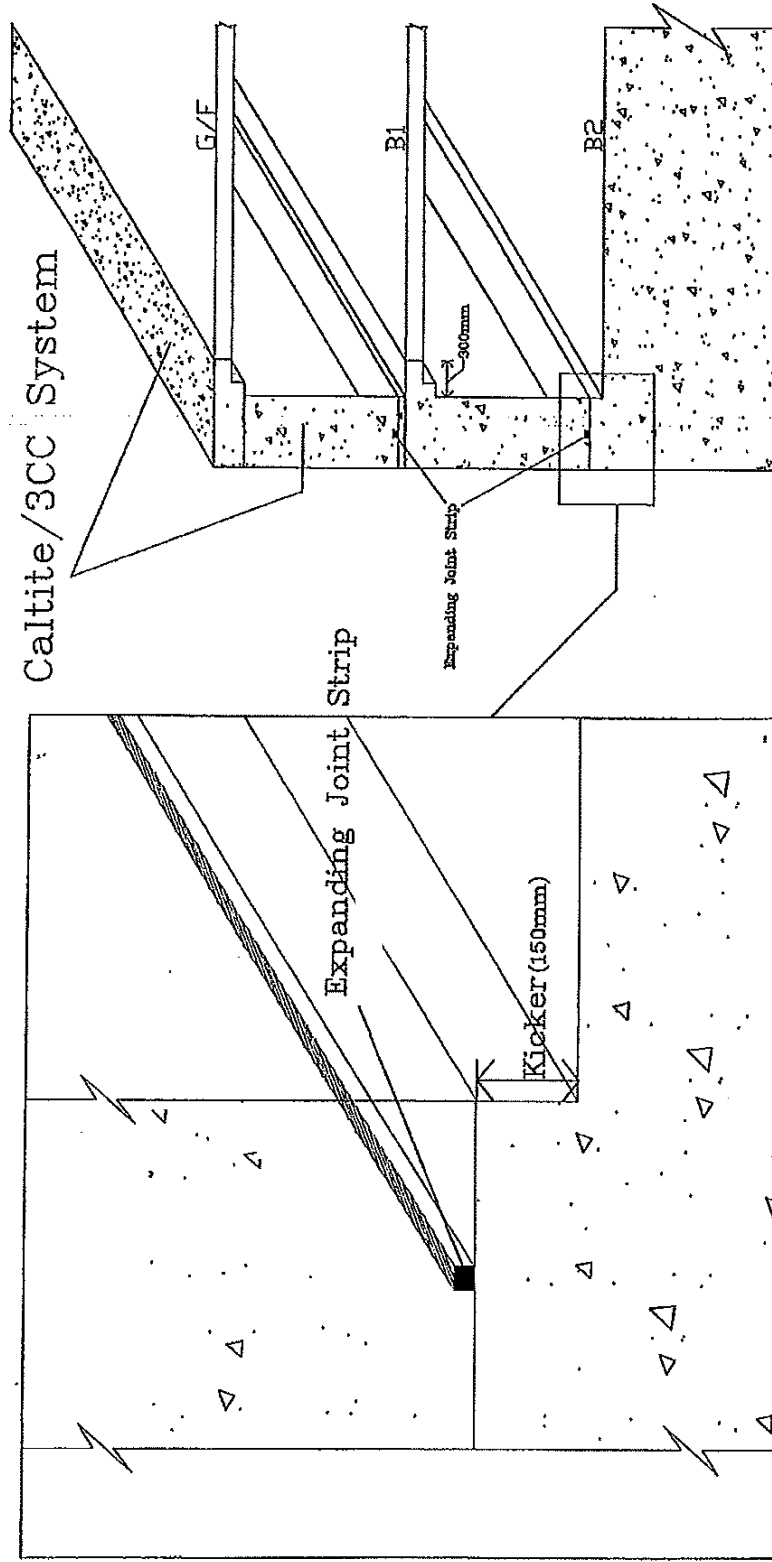


Fig B : Kicker Joints Detail (Slabs)

1. At all wall / floor joints, the floors are to have a minimum 150mm high wall 'kicker' cast in place at the slab edges.
2. A Cementaid Expanding Joint Strip is to be cast in place at all construction joint locations.
3. All hardened concrete construction joint surfaces are to be roughened, cleaned, and free from loose particles or dust to allow optimum bond.
4. The basement slab vertical construction joint is to be liberally thoroughly brushed with Cementaid **WELDCRETE** Bonding Agent / cement slurry. The **WELDCRETE** must be wet for optimum bond, do not allow the **WELDCRETE** to dry or become tacky or bond failure will occur. In the case of drying out, apply fresh **WELDCRETE** Slurry before continuing.
5. The wall concrete is to be placed directly onto or against the prepared joint and thoroughly compacted.

NOTE: For vertical joints, the forming material is to be pre-treated with CEMENTAID **CALSTRIP** form release and keying agent. After stripping, all loose material and retarded mortar is to be thoroughly washed and brushed away.

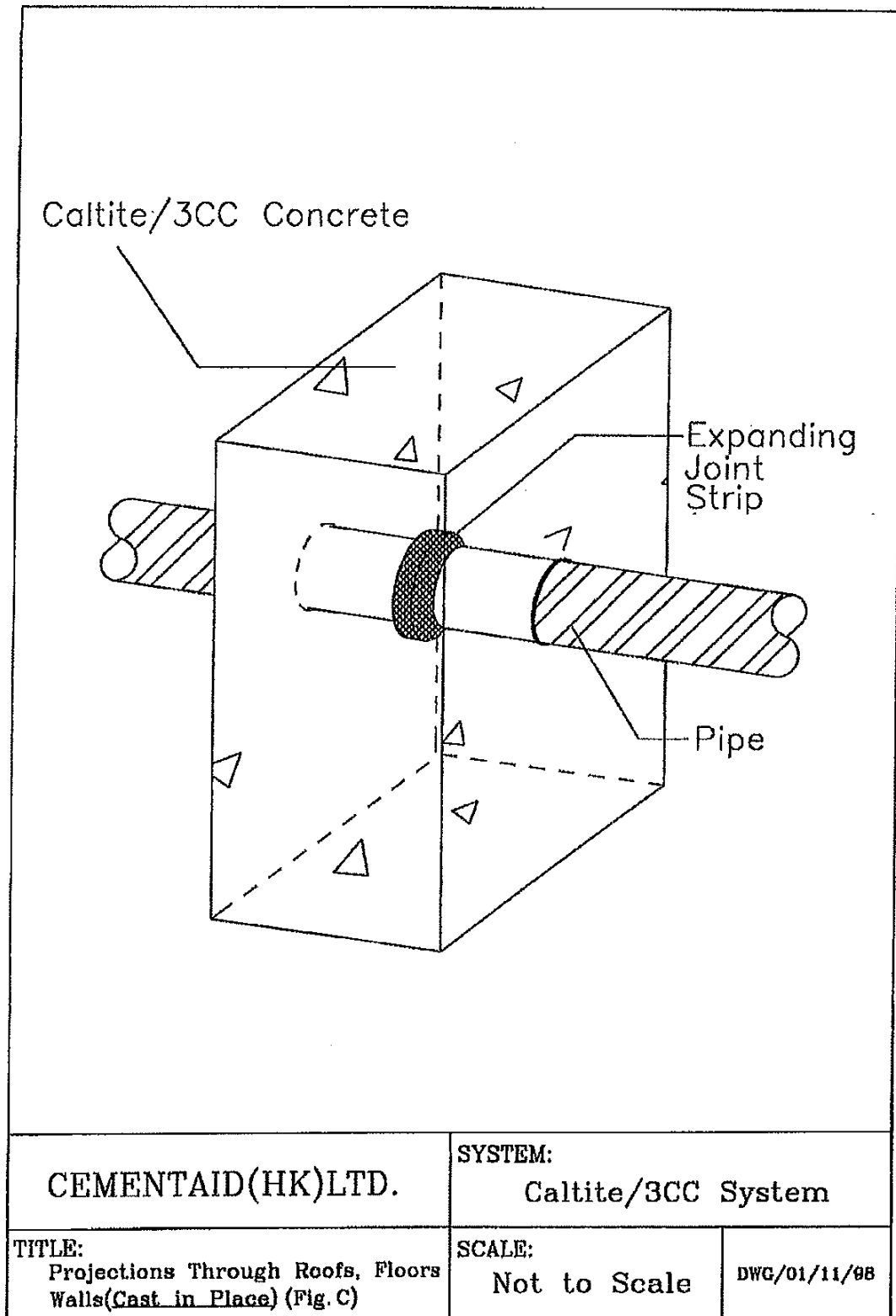


CEMENTAID (HK) LTD.	TITLE: CJ of Basement Slab-to-Wall (Fig.B)	SYSTEM: Caltite/3CC System	
		Scale: Not to Scale	DWG/10/04/12

**Fig C : Projection Through Roof, Floors and Walls
 (Cast in Place)**

1. The CEMENTAID EXPANDING JOINT STRIP is secured around the middle of projection.
2. EVERDURE CALTITE / 3CC concrete is simply cast around the projection and thoroughly compacted. It is important to seal all formwork around the projection so as to avoid grout loss.

NOTE: It is also possible to use pre-formed sleeve waterstops where these are available.

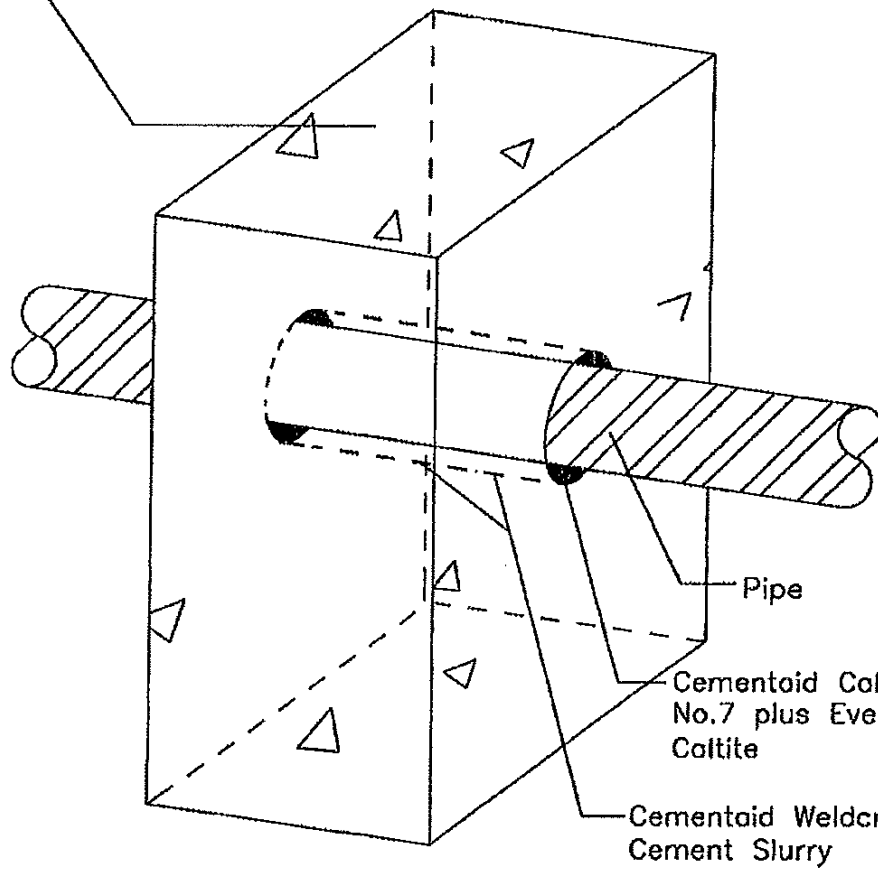


**Fig D : Projections Through Roof, Floors and Walls
 (Through Prepared Holes)**

1. The edges of the prepared hole are to be thoroughly cleaned of all loose debris, dust and contaminants.
2. The projection is to be positioned correctly through the hole.
3. All surfaces of the concrete and projection are to be thoroughly brushed with Cementaid **WELDCRETE** / cement slurry and the space between the **EVERDURE CALTITE** / 3CC concrete & the projection filled with Cementaid **CALGROUT NO.7** - Non Shrink Grout, containing **EVERDURE CALTITE SYSTEM** / 3CC **SYSTEM**, and finished to the concrete level.

NOTE : The **WELDCRETE** must be used while **WELDCRETE** is wet, not dry or tacky. If **WELDCRETE** dries out it is to be reapplied before continuing with the grouting.

Caltite/3CC Concrete



CEMENTAID(HK)LTD.

SYSTEM:

Caltite/3CC System

TITLE:

Projections Through Roofs, Floors
& Walls(Through Prepared Holes)(FigD)

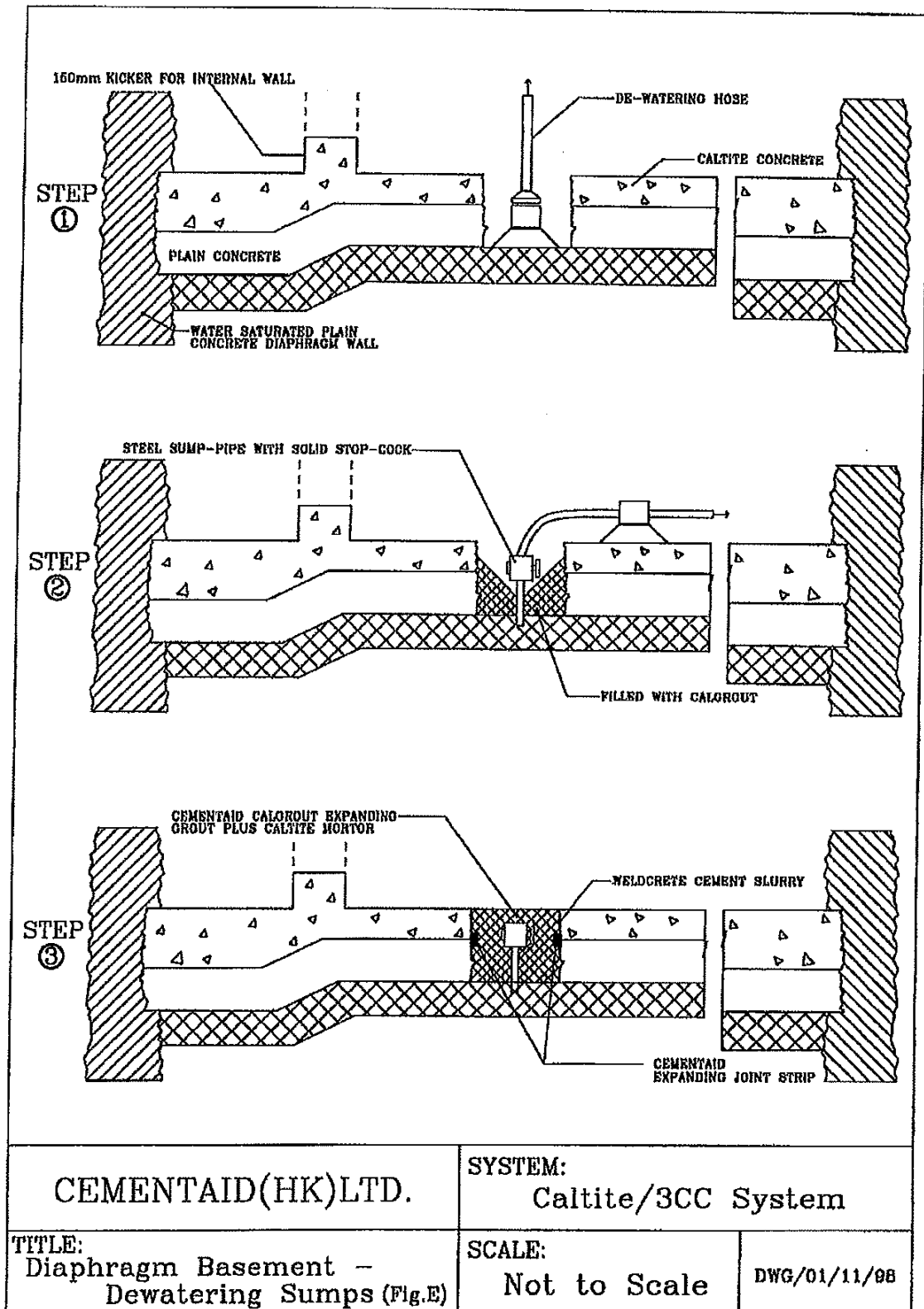
SCALE:

Not to Scale

DWG/01/11/98

Fig. E : HOLES THROUGH GROUND SLABS BELOW WATER

TABLE DIAPHRAGM BASEMENT – DEWATERING SUMPS



NOTE : Detail above, shows **CEMENTAID EVERDURE CALTITE** concrete placed monolithically as top 400mm of 800mm basement slab. Lower slab portion is plain concrete. This type of “split/homogeneous” raft, or pile cap, construction is quite common and easily constructed at optimum cost savings.

- 1.0
 - Form correct sized holes in ground floor slab – use **CEMENTAID CALSTRIP** to achieve good, roughened, ‘keyed’ finish from hole forms.
 - Install sump pump and de-water before placing surrounding slab.
- 2.0
 - Remove sump pump.
 - Install steel sump pipe with solid stop-cock type tap fitting at top end. Approx. level with beginning of **EVERDURE CALTITE** concrete. Top of pipe should be around 150mm from top of **EVERDURE CALTITE** slab surface.
 - Attach pump to top of steel sump pump to continue de-watering
 - Pack sump pipe in place using **CEMENTAID CALGROUT**, expanding grout.
 - Allow 3 days for sufficient strength gain at least.
- 2.A **TEST.** Turn off stop-cock tap for 2 hours at least, and check **CALGROUT** ‘plug’ for any leaks. If OK, resume de-watering. If leaks evident, hack out localized leaks and re-grout.
- 3.0
 - Apply expanding joint strip to all sides of de-watering sump hole approx. 150mm from top of **EVERDURE CALTITE** slab.
 - Turn off stop-cock tap, and disconnect pump hose.
 - Blow out water from top of sump pipe and sump hole.
 - Prime all surfaces with **CEMENTAID WELDCRETE**/cement slurry, 1:1 (Do not coat joint strip)
 - Pack top of sump pipe, and sump hole to slab surface level, using **CEMENTAID CALGROUT** expanding grout containing **CEMENTAID EVERDURE CALTITE**. Cure properly.

Fig E Internal floor slab. 'Perimeter waterproofing' for Diaphragm Wall/"Wet-Wall" basement constructions.

Where Diaphragm Wall ("Wet-Wall") construction is used in deep basements, use of impermeable **EVERDURE CALTITE** concrete in the thick walls themselves may not be economically justified (N.B. Economically sound when maximum, usable internal floor area is at a premium – high rent shops, banks, hotels, ect.) Where this is the case, normal internal suspended floors are butt-jointed into the Wet-Wall and an internal drainage channel and wall system is used. This collects water and moisture seepage coming through the Diaphragm Wall and channels it to sump for disposal. However, normal, absorptive plain concrete, in contact with the wet external wall, rapidly promotes 'wick' action of water throughout the internal slab. **EVERDURE CALTITE** external structural walls do not have this problem.

To permanently, easily and effectively keep internal concrete floors dry and habitable/usable it is necessary to cast a 1 meter perimeter of the floor in impermeable **EVERDURE CALTITE** concrete. The suggested detail below demonstrates this. The **EVERDURE CALTITE** perimeter can be placed separately to or monolithically with, the remaining plain concrete slab. Note that the seamless, cast-in-place **EVERDURE CALTITE** perimeter and 'kicker', ensures drain channel integrity, too!

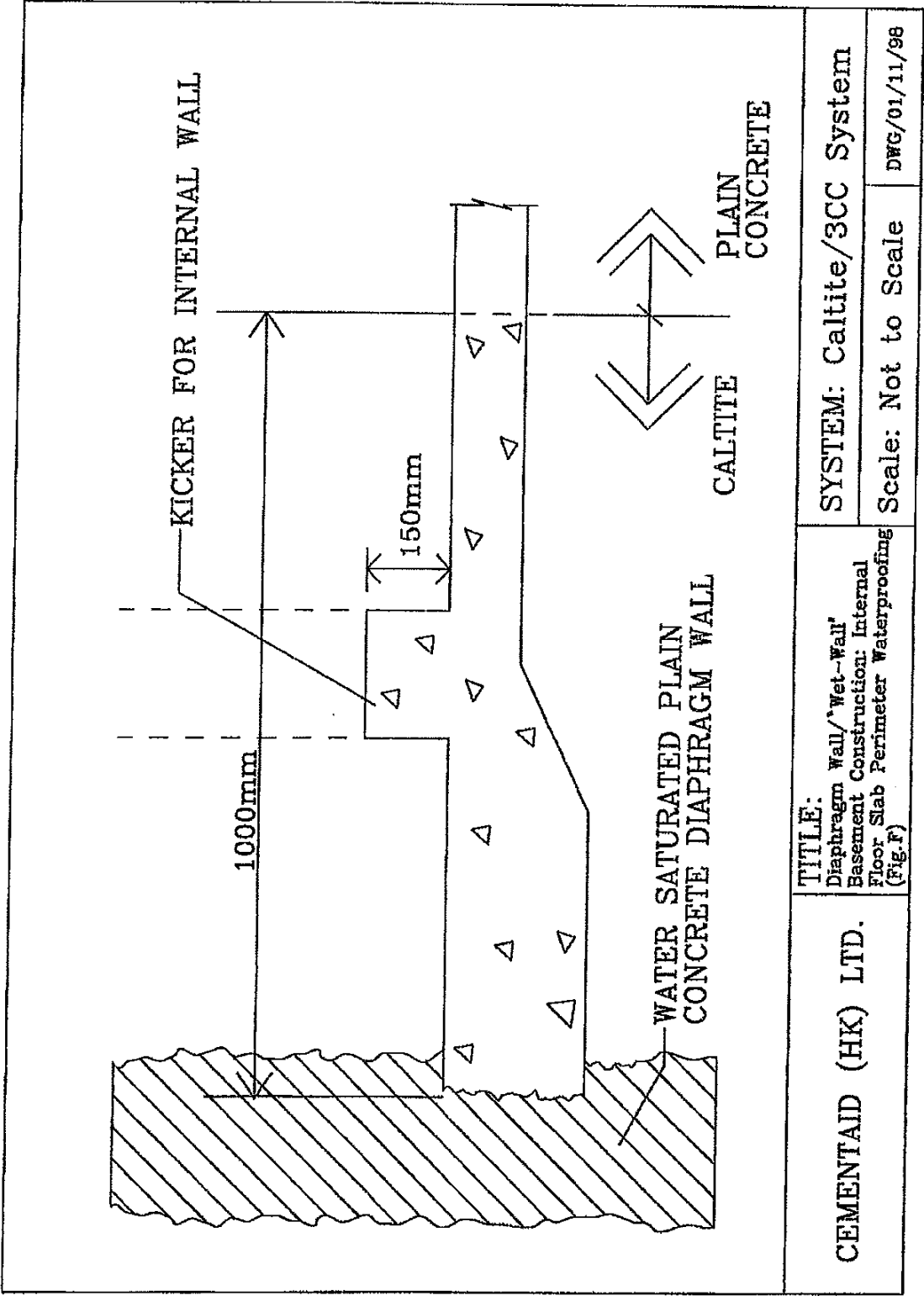
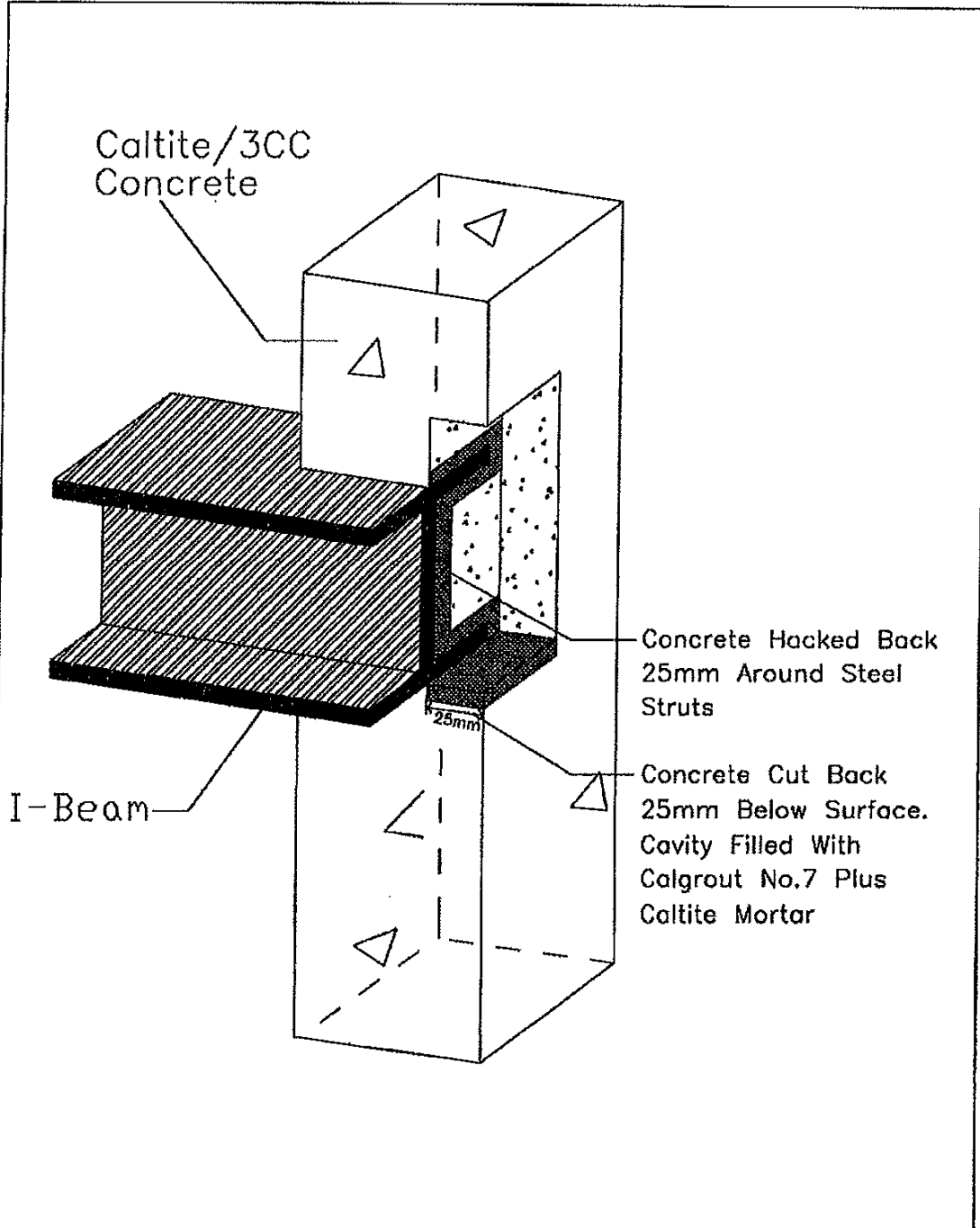


Fig G. GROUTING FOR SHORING STRUTS, TIE RODS, ETC.

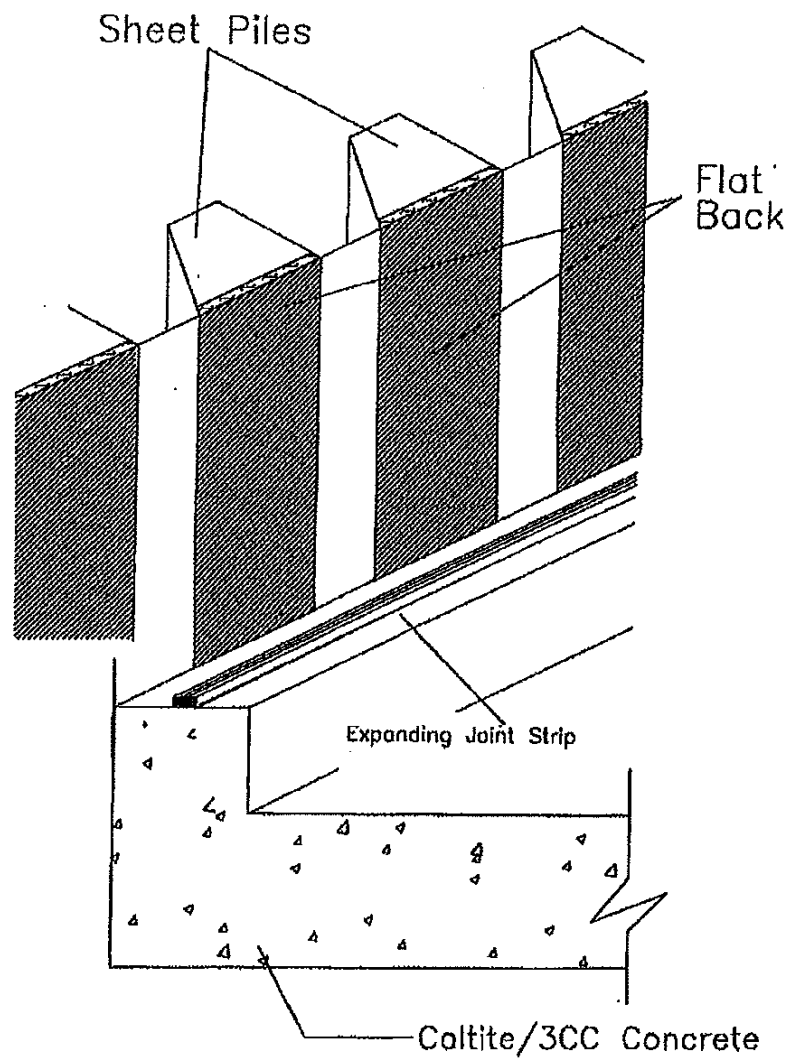
- Where Shoring Struts, Tie Rods, or other metal is to be left embedded in a structural **EDC** concrete wall :

Method

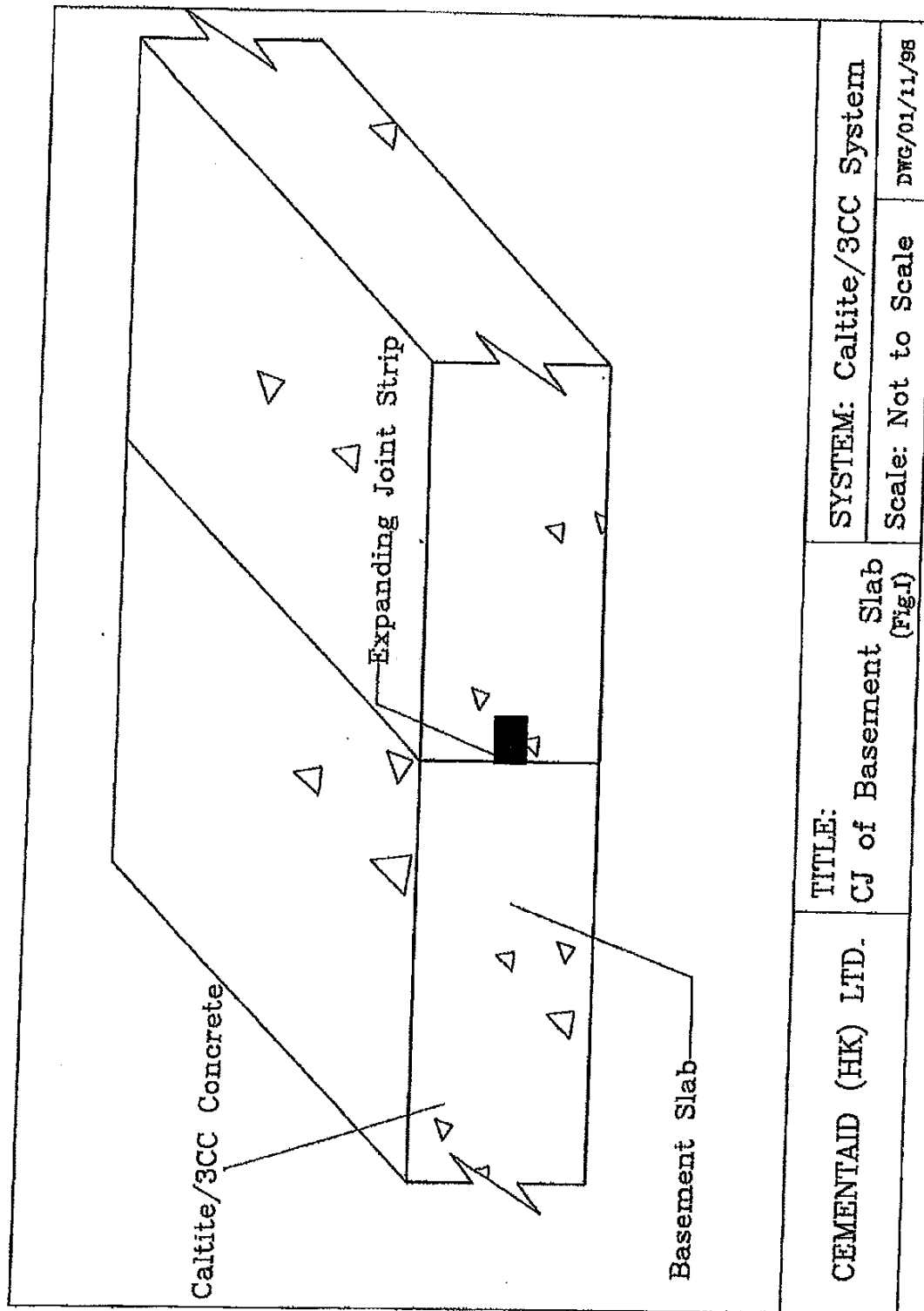
1. Hack back concrete to minimum 25mm around steel.
2. Cut back steel to minimum 25mm below concrete surface.
3. Clean cavity to free from all loose particles and contaminants.
4. Check whether there is any visible water coming from around the steel. If there is water leakage, carry out further hacking of concrete to additional 25mm around steel. Stop the running water by packing with **CEMENTAID CALFLASH** flash-setting grout.
5. When water leakage is stopped, dampen surface of cavity with water and apply **CEMENTAID WELDCRETE** / cement slurry (1:1) as bonding coat.
6. While the bonding coat is still wet fill cavity with **CEMENTAID CALGROUT NO.7** with **CALTITE** added (Dose : 5 litres of **CALTITE** per 100kg of grout).



CEMENTAID(HK)LTD.	SYSTEM: Caltite/3CC System	
TITLE: Grouting for Shoring Struts, TieRod, etc. (Fig.G)	SCALE: Not to Scale	DWG/01/11/98



CEMENTAID(HK)LTD.	SYSTEM: Caltite/3CC System	
TITLE: CJ of Slab to Wall with SheetPiling (Fig. H)	SCALE: Not to Scale	DWG/01/11/98



CEMENTAID (HK) LTD.	TITLE: CJ of Basement Slab (Fig.1)	SYSTEM: Caltite/3CC System	
		Scale: Not to Scale	DWG/01/11/98

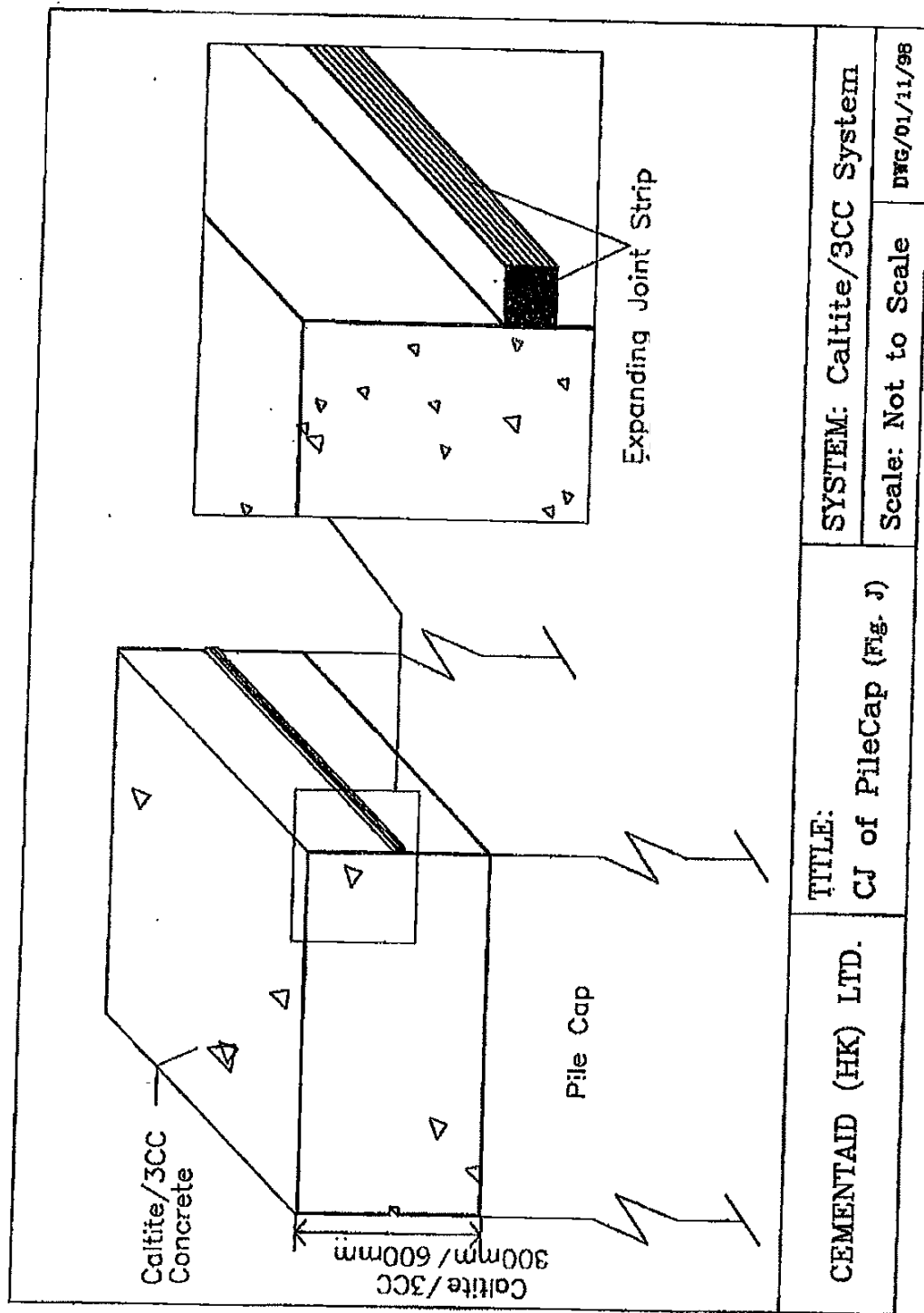


Fig K. RE-ENTRANT ANGELES

Where re-entrant angles occur, three 10mm or 12mm h.r.d. bars, spaced 100mm apart, must be placed top and bottom. At 90 degrees across the angles. i.e. Light Box, Shoring Strut, plinth and any square opening etc.

